

Tarea dección 7.2

$$1) \int \sin^3 x \cos^2 x dx = \int \sin^2 x \sin x \cos^2 x dx = \int (1 - \cos^2 x) \sin x \cos^2 x dx$$

$$\begin{aligned} u &= \cos x \\ du &= -\sin x dx \\ &= \int (1 - u^2) \sin x u^2 \frac{du}{-\sin x} = - \int (1 - u^2) u^2 du \\ &= - \int u^2 - u^4 du = - \left(\frac{u^3}{3} + \frac{u^5}{5} \right) = \frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} + C \end{aligned}$$

$$3) \int_0^{\pi/2} \sin^7 \theta \cos^5 \theta d\theta = \int \sin^6 \theta \sin \theta \cos^5 \theta d\theta = \int (\sin^2 \theta)^3 \sin \theta \cos^5 \theta d\theta$$

$$= \int (1 - \cos^2 \theta)^3 \sin \theta \cos^5 \theta d\theta = \int (1 - u^2)^3 \sin \theta u^5 \frac{du}{-\sin \theta}$$

$$\begin{aligned} &= - \int (1 - u^2)^3 u^5 du = - \int (-u^6 + 3u^4 - 3u^2 + 1) u^5 du = - \int -u^11 + 3u^9 - 3u^7 + u^5 du \\ &= - \left(-\frac{u^{12}}{12} + \frac{3u^{10}}{10} - \frac{3u^8}{8} + \frac{u^6}{6} \right) = \frac{\cos^{12} \theta}{12} - \frac{3\cos^{10} \theta}{10} + \frac{3\cos^8 \theta}{8} - \frac{\cos^6 \theta}{6} \Big|_0^{\pi/2} \\ &= \frac{1}{120} \end{aligned}$$

$$5) \int \sin^5(2t) \cos^2(2t) dt = \int \sin^4 u \cos^2 u \frac{du}{2} = \frac{1}{2} \int \sin^4 u \cos^2 u du$$

$$\begin{aligned} u &= 2t \\ du &= 2 dt \\ &= \frac{1}{2} \int \sin^4 u \sin u \cos^2 u du = \frac{1}{2} \int (\sin^2 u)^2 \sin u \cos^2 u du = \end{aligned}$$

$$= \frac{1}{2} \int (1 - \cos^2 u)^2 \sin u \cos^2 u du = \frac{1}{2} \int (1 - m^2)^2 \sin u m^2 \frac{dm}{-\sin u} = -\frac{1}{2} \int (1 - m^2)^2 m^2 dm$$

$$\begin{aligned} m &= \cos u \\ dm &= -\sin u du \\ &= -\frac{1}{2} \int (1 - 2m^2 + m^4) m^2 dm = -\frac{1}{2} \int (m^2 - 2m^4 + m^6) dm \\ &= -\frac{1}{2} \left(\frac{m^3}{3} - \frac{2m^5}{5} + \frac{m^7}{7} \right) = -\frac{1}{6} m^3 + \frac{1}{5} m^5 - \frac{1}{14} m^7 \end{aligned}$$

$$= -\frac{\cos^3 u}{6} + \frac{\cos^5 u}{5} - \frac{\cos^7 u}{14} = -\frac{\cos^3(2t)}{6} + \frac{\cos^5(2t)}{5} - \frac{\cos^7(2t)}{14} + C$$

$$7) \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = \int \frac{1 + \cos 2\theta}{2} d\theta = \frac{1}{2} \int 1 + \cos 2\theta d\theta$$

$$= \frac{1}{2} \left[\int 1 d\theta + \int \cos 2\theta d\theta \right] = \frac{1}{2} \left[\theta + \int \cos u \frac{du}{2} \right] = \frac{1}{2} \left[\theta + \frac{1}{2} \int \cos u du \right]$$

$$u = 2\theta \quad du = 2d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{1}{2} \sin \theta \right] \Big|_0^{\frac{\pi}{2}} = \frac{1}{2} \left(\frac{\pi}{2} + 0 \right) = \frac{\pi}{4}$$

$$9) \int_0^{\pi} \cos^4(2t) dt = \int \cos^4(u) \frac{du}{2} = \frac{1}{2} \int \cos^3 u \cos u du =$$

$$u = 2t \quad du = 2dt$$

$$= \cos^3 u \sin u - \int \sin u - 3 \cos^2 u \sin u du =$$

$$= \cos^3 u \sin u + 3 \int \sin u \cos^2 u \sin u du =$$

$$= \cos^3 u \sin u + 3 \int \sin^3 u \cos^2 u du =$$

$$u = \cos^3 u$$

$$du = -3 \cos^2 u \sin u du$$

$$dv = \cos u du$$

$$v = \sin u$$

$$= \cos^3 u \sin u + 3 \int \cos^2 u du$$

$$= \cos^3 u \sin u + 3 \int \cos m \frac{dm}{2}$$

$$= \cos^3 u \sin u + \frac{3}{2} \int \cos m dm$$

$$= \cos^3 u \sin u + \frac{3}{2} \sin m \Big|_0^{\pi}$$

$$= \cos^3 2t \sin 2t + \frac{3}{2} \sin 4t \Big|_0^{\pi}$$

$$71) \int_0^{\frac{\pi}{2}} \sin^2 x \cos^2 x dx = \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right) dx = \int \frac{1 - \cos 4x}{8} dx$$

$$\frac{1}{8} \int 1 - \cos 4x dx = \frac{1}{8} \int 1 - \cos u \frac{du}{4} = \frac{1}{8} \left[\int 1 du - \frac{1}{4} \int \cos u du \right]$$

$$u = 4x \quad du = 4dx$$

$$= \frac{1}{8} \left[u - \frac{1}{4} \sin u \right] = \frac{1}{8} \left[u \Big|_0^{\frac{\pi}{2}} - \frac{1}{4} \sin u \Big|_0^{\frac{\pi}{2}} \right]$$

$$= \frac{1}{8} \left[\frac{\pi}{2} - 0 \right] = \frac{\pi}{16}$$

$$13) \int \sqrt{\cos \theta} \sin^3 \theta d\theta = \int \sqrt{\cos \theta} \sin^2 \theta \sin \theta d\theta = \int \sqrt{u} \sin^2 \theta \frac{du}{- \sin \theta}$$

$$u = \cos \theta \quad \therefore -\int \sqrt{u} \sin^2 \theta du = -\int \sqrt{u} (1 - \cos^2 \theta) du$$

$$du = -\sin \theta d\theta \quad = -\int u^{1/2} (1 - u^2) du = -\int u^{1/2} - u^{5/2} du =$$

$$= -\frac{2}{3} u^{3/2} + \frac{2}{7} u^{7/2} = -\frac{2}{3} \cos^{3/2} \theta + \frac{2}{7} \cos^{7/2} \theta + C$$

$$15) \int \cot x \cos^2 x dx = \int \left(\frac{\cos x}{\sin x} \right) \cos^2 x dx = \int \frac{\cos x \cos^2 x}{\sin x} dx$$

$$= \int \frac{(1 - \sin^2 x) \cos x}{\sin x} dx = \int \frac{(1 - u^2) \cos x}{u} \frac{du}{\cos x} = \int \frac{1 - u^2}{u} du = \int \frac{1}{u} - u du$$

$$u = \sin x \quad = (\ln|u| - \frac{u^2}{2}) = (\ln|\sin x| - \frac{1}{2} \sin^2 x) + C$$

$$du = \cos x dx$$

$$17) \int \sin^3 x \cos^2 x dx = \int \frac{1 - \cos^2 x}{2} \sin^2 x dx = \frac{1}{2} \int (1 - \cos^2 x) \sin^2 x dx$$

$$u = 1 - \cos^2 x \quad = \frac{1}{2} \left[\int u \sin^2 x \frac{du}{2 \sin x} \right] = \frac{1}{4} \int u du = \frac{1}{4} \cdot \frac{u^2}{2}$$

$$du = \sin(2x) 2 dx$$

$$= \frac{1}{4} \cdot \frac{1 - \cos^2 x}{2} = \frac{1}{8} (1 - \cos^2 x)^2 = \frac{1}{8} \sin^4 x + C$$

$$19) \int t \sin^2 t dt = \frac{1}{2} t \left[1 - \frac{1}{2} \sin 2t \right] - \frac{1}{2} \int \left[t - \frac{1}{2} \sin 2t \right] dt = \frac{1}{2} \left[\int t dt - \frac{1}{2} \int \sin 2t dt \right]$$

$$u = t \quad dv = \sin^2 t dt \quad n = 2t$$

$$du = dt \quad v = \int \sin^2 t dt = \int \frac{1 - \cos 2t}{2} dt \quad dn = 2 dt$$

$$= \frac{1}{2} \int 1 - \cos 2t dt$$

$$= \frac{1}{2} \left[\int 1 dt - \int \cos 2t dt \right]$$

$$= \frac{1}{2} \left[\int 1 dt - \int \cos m \frac{dm}{2} \right]$$

$$= \frac{1}{2} \left[\frac{t^2}{2} - \frac{1}{2} \int \sin n \frac{dn}{2} \right]$$

$$= -\frac{1}{2} \left[\frac{t^2}{2} - \frac{1}{4} \int \sin n dn \right]$$

$$= -\frac{1}{2} \left[\frac{t^2}{2} + \frac{1}{4} \cos 2t \right]$$

$$= \frac{1}{2} t \left[t - \frac{1}{2} \sin 2t \right] - \frac{1}{2} \left[\frac{t^2}{2} + \frac{1}{4} \cos 2t \right] + C$$

$$m = 2t \quad = \frac{1}{2} \left[\int 1 dt - \frac{1}{2} \int \cos m dm \right]$$

$$dm = 2 dt \quad = \frac{1}{2} \left[t + \frac{1}{2} \sin 2t \right]$$

$$21) \int \tan x \sec^3 x dx = \int \tan x \cdot u \cdot \frac{du}{3 \sec^2 x \tan x}$$

$$u = \sec^3 x$$

$$du = 3 \sec^2 x \sec x \tan x$$

$$du = 3 \sec^2 x \tan x$$

$$= \int \tan x \cdot u \cdot \frac{du}{3 u \tan x}$$

$$= \frac{1}{3} \int du = \frac{1}{3} u = \frac{1}{3} \sec^3 x + C$$

$$23) \int \tan^2 x dx = \int \sec^2 x - 1 dx = \int \sec^2 x dx - \int 1 dx = \tan x - x + C$$

$$25) \int \tan^4 x \sec^6 x dx = \int \tan^4 x (\sec^2 x)^3 \sec^2 x dx$$

$$= \int \tan^4 x (\tan^2 x + 1)^2 \sec^2 x dx = \int u^4 (u^2 + 1)^2 \sec^2 x \frac{du}{\sec^2 x}$$

$$u = \tan x$$

$$du = \sec^2 x$$

$$= \int u^4 (u^2 + 1)^2 du = \int u^4 (u^4 + 2u^2 + 1) du$$

$$= \int u^8 + 2u^6 + u^4 du = \frac{u^9}{9} + \frac{2u^7}{7} + \frac{u^5}{5}$$

$$= \frac{1}{9} \tan^9 x + \frac{2}{7} \tan^7 x + \frac{1}{5} \tan^5 x + C$$

$$27) \int \tan^3 x \sec x dx = \int \tan^2 x \tan x \sec x dx = \int (\sec^2 x - 1) \tan x \sec x dx$$

$$u = \sec x$$

$$du = \sec x \tan x$$

$$= \int (u^2 - 1) \tan x \cdot u \cdot \frac{du}{\sec x \tan x} = \int (u^2 - 1) \tan x \cdot u \cdot \frac{du}{u \tan x}$$

$$= \int \frac{u(u^2 - 1)}{u} du = \int u^2 - 1 du$$

$$= \frac{u^3}{3} - u = \frac{1}{3} \sec^3 x - \sec x + C$$

$$29) \int \tan^3 x \sec^6 x dx = \int \tan^2 x \sec^2 x (\sec^2 x)^2 dx = \int \tan^2 x \sec^2 x (\tan^2 x + 1)^2 dx$$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$= \int u^2 \sec^2 x (u^2 + 1)^2 \frac{du}{\sec^2 x} = \int u^2 (u^2 + 1)^2 du = \int u^2 (u^4 + 2u^2 + 1) du$$

$$= \int u^6 + 2u^4 + u^2 = \frac{1}{7} u^7 + \frac{2}{5} u^5 + \frac{1}{3} u^3 = \frac{\tan^7 x}{7} + \frac{2 \tan^5 x}{5} + \frac{\tan^3 x}{3} + C$$

$$31) \int \tan^5 x dx = \int \tan^4 x \tan x dx = \int (\sec^2 x - 1)^2 \tan x = \int (u^2 - 1)^2 \tan x \frac{du}{\sec^2 x \tan x}$$

$$u = \sec x$$

$$du = \sec x \tan x$$

$$= \int (u^2 - 1)^2 \tan x \frac{du}{u \tan x} = \int \frac{(u^2 - 1)^2}{u} du = \int \frac{u^4 - 2u^2 + 1}{u} du$$

$$= \int \frac{u^4}{u} - \frac{2u^2}{u} + \frac{1}{u} du = \int u^3 - 2u + \frac{1}{u} du = \frac{u^4}{4} - u^2 + \ln|u|$$

$$= \frac{1}{4} \sec^4 x - \sec^2 x + \ln|\sec x| + C$$

$$33) \int x \sec x \tan x dx = x \sec x - \int \sec x dx = x \sec x - \ln|\tan x + \sec x| + C$$

$$u = x$$

$$dv = \sec x \tan x dx$$

$$du = dx$$

$$v = \int \sec x \tan x dx$$

$$v = \sec x$$

$$35) \int_{\pi/4}^{\pi/2} \cot^2 x dx = \int \csc^2 x - 1 = \int \csc^2 x dx - \int 1 dx = -\cot x - x + C$$

$$37) \int_{\pi/4}^{\pi/2} \cot^5 \theta \csc^3 \theta d\theta = \int \cot^4 \theta \underbrace{\cot \theta \csc^3 \theta}_{-d\csc \theta} = -\int \cot^4 \theta \csc^2 \theta d\csc \theta$$

$$u = \csc \theta$$

$$du = -\cot \csc \theta d\theta$$

$$= \int \cot^4 \theta u^2 du =$$

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$$39) \int \sec x = \int \frac{1+u^2}{2u} \cdot \frac{2du}{1+u^2} = \int \frac{1}{u} du$$

$$u = \tan \frac{x}{2} \quad = \ln|u| = \ln|\tan \frac{x}{2}| + C$$

$$du = \frac{1}{1+u^2} dx$$

$$41) \int \sin 8x \cos 5x dx = \int \frac{\sin(8x+5x) + \sin(8x-5x)}{2} dx$$

$$= \frac{1}{2} \int \sin(13x) + \sin(3x) = \frac{1}{2} [-\cos 13x - \cos 3x]$$

$$u = 13x \quad m = 3x \quad = \frac{1}{2} \left(-\frac{1}{13} \cos 13x - \frac{1}{3} \cos 3x \right) + C$$

$$du = 13dx \quad dm = 3dx$$

$$43) \int_0^{\pi/2} \cos 5t \cos 7t dt = \frac{1}{2} \int (\cos(5t+7t) + \cos(5t-7t)) dt$$

$$= \frac{1}{2} \int (\cos(12t) + \cos(-2t)) dt$$

$$0 = 5t \quad m = 7t \quad \frac{1}{2} [\sin 12t + \sin 5t]$$

$$du = 12dt \quad dm = 5dt \quad \frac{1}{2} \left[\frac{1}{12} \sin 12t + \frac{1}{5} \sin 5t \right] \Big|_0^{\pi/2} =$$

$$= \frac{1}{2} \left(\frac{1}{5} - \frac{1}{12} \right) = \frac{1}{120}$$

$$45) \int_0^{\pi/6} \sqrt{1+\cos 2x} dx = \int \sqrt{1-1+2\cos^2 x} dx = \int \sqrt{2} \sqrt{\cos^2 x} dx$$

$$= \sqrt{2} \int \cos x = \sqrt{2} \sin x \Big|_0^{\pi/6} = \sqrt{2} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}}$$

$$47) \int \frac{1 - \tan^2 x}{\sec^2 x} dx = \int 1 - \frac{\left(\frac{\sin x}{\cos x}\right)^2}{\frac{1}{\cos^2 x}} dx = \int 1 - \frac{\sin^2 x}{\cos^2 x} dx$$

$$= \int \frac{\cos^2 x - \sin^2 x}{\cos^2 x} dx = \int \cos^2 x - \sin^2 x dx = \int \cos 2x dx$$

$$u = 2x \\ du = 2 dx$$

$$= \int \cos u \frac{du}{2} = \frac{1}{2} \int \cos u = \frac{1}{2} \sin 2x + C$$

$$48) \int x \tan^2 x dx = x(\tan x - x) - \int (\tan x - x) dx$$

$$u = x \\ du = dx$$

$$\begin{aligned} dv &= \tan^2 x dx \\ v &= \int \tan^2 x dx \\ v &= \int \sec^2 x - 1 dx \\ v &= \tan x - x \end{aligned}$$

$$\int = \frac{x^2}{2} - \ln |\cos x|$$

$$= x^2 + x \tan x + \frac{x^2}{2} + \ln |\cos x| + C$$